

20620 exam 2024F solutions:

Q1

Solution: Na, Al, P, Cl.

In a period, the atomic radius decreases from left to the right. Thus, Na, Al, P and Cl are placed from left to right in the period 3. Alternatively, Figure 8.4 in Chapter 8 in the text book also provides the actual atomic radius of the listed elements.

Atomic radii: Na (186 pm), Al (143 pm), P (110 pm) and Cl (99 pm)

Q2

Solution: ${}^{37}_{17}\text{Cl}$

Cl has the atomic number of 17. The atomic mass given in the question for Cl isotope is 37.

The acceptable form to denote the atomic mass and atomic number of an atom of an element, in this case Cl, is ${}^{37}_{17}\text{Cl}$

Q3

Solution: 11.1 mol, 511 g

Fermentation reaction given: $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow \text{CO}_2 + \text{CH}_3\text{CH}_2\text{OH}$

Balance the reaction: $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{CO}_2 + 2\text{CH}_3\text{CH}_2\text{OH}$

Mass of Fructose: 1000 g. Molar mass of Fructose: 180 g/mol. Mol of Fructose: $1000/180 = 5.56$ mol. Based on the balanced fermentation equation – 1 mol of fructose results in 2 mol of ethanol. The number of mol of ethanol produced: $2 \times 5.56 = 11.12$ mol. The molar mass of ethanol: 46 g/mol. The mass of ethanol produced: $46 \times 11.12 = 511.52$ g

Q4

Solution: 9.24 MPa.

$\text{H}_2 + 0.5 \text{O}_2 \rightarrow \text{H}_2\text{O}$. Based on the given masses of H_2 and O_2 , the mol of hydrogen are 5 and the mol of oxygen are 0.5. Oxygen is the limiting reactant. 0.5 mol of oxygen react with 1 mol of hydrogen to produce 1 mol of water and leaving 4 mol H_2 . Thus, 5 mol of gas is in the vessel.

Using the ideal gas equation: $PV=nRT \Rightarrow V=nRT/P$:

$V = 10 \text{ L} = 0.01 \text{ m}^3$, $n = 5 \text{ mol}$, $T = 1950 + 273 = 2223 \text{ K}$ and $R = 8.314 \text{ J/mol}\times\text{K}$

$P = (5 \times 8.314 \text{ J/mol}\times\text{K} \times 2223 \text{ K})/0.01 \text{ m}^3 = 9.24 \text{ MPa}$.

Q5

Solution: -1138 kJ/mol

$$\Delta G^0 = \Delta H^0 - T\Delta S^0 \Rightarrow \Delta H^0 = \Delta G^0 + T\Delta S^0$$
$$\Delta H^0 = -1080200 \frac{\text{J}}{\text{mol}} + 323.12 \text{ K} \times \left(-180 \frac{\text{J}}{\text{Kmol}}\right) = -1138367 \frac{\text{J}}{\text{mol}} = -1138 \frac{\text{kJ}}{\text{mol}}$$

Q6

Solution: The reaction is not spontaneous at low temperature but spontaneous at high temperature.

When both ΔH and ΔS are positive, then T has to be large enough to make ΔG negative (so that $T\Delta S$ is larger than ΔH and results in negative ΔG).

Thus, at low T , ΔG becomes positive making the reaction non-spontaneous, while at high temperatures ΔG becomes negative making the reaction spontaneous.

Q7

Solution: $n = 4, l = 2, m = 0$

The electron configuration of iodine is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^5$

The electron in $4d$ is in consideration. The value of n is 4 for $4d$ orbital. The value of l is 2 for d orbital. The value of m varies between -2, -1, 0, 1, 2 for d orbitals.

With the above combinations, among the given options, only $n = 4, l = 2, m = 0$ is feasible.

Q8

Solution: 2, 8, 8, 8, 8

Looking at the electron configurations for the listed noble gas atoms, the outer shell electron configurations are:

He: $1s^2$

Ne: $2s^2 2p^6$

Ar: $3s^2 3p^6$

Kr: $4s^2 4p^6$

Xe: $5s^2 5p^6$

Q9

Solution: Only BH_3

NH_3 has the molecular geometry of trigonal pyramidal – 3 bonds and one lone pair.

SF_6 has the geometry of octahedral – 6 bonds and no lone pairs

BH_3 has the geometry of trigonal planar – 3 bonds and no lone pair

CH_4 has the geometry of tetrahedral – 4 bonds and no lone pair

Q10

Solution: Intermolecular bonds

In solid argon, the atoms are held together by weak van der Waals forces.

Q11

12 g of NaBr contain $2 \times m_{NaBr}/M_{NaBr}$ mol of particles. This is dissolved in 300 ml = 0.3 kg water. Thus the freezing point depression is

$$\Delta T_{freezing\ point} = K_f 2 \times \frac{m_{NaBr}}{M_{NaBr}} = -2 \frac{1.86^\circ C}{m} \times \frac{12g}{0.3kg \times 102.89g/mol} = -1.45^\circ C$$

Q12

The BCC unit cell contains 2 atoms. The volume of a cube with two atoms is $\frac{2m_{Fe}}{\rho_{Fe}} = \frac{\frac{2M_{Fe}}{N_A}}{\rho_{Fe}} =$

$$2 \times \frac{\frac{55.85g}{mol} \times \frac{10^{23}}{6.02 \times 10^{23}}}{\frac{7874kg}{m^3}} = 2.36 \times 10^{-29} m^3.$$

Q13

You can either realize that the concentration is cut in half from both 0 to 60 s and from 60 s to 120 s. You can then check if then distance (in time from 120 s to 300 s – 180 s) lead to a decline in concentration of $\frac{1}{2}^3$. This is the case thus $\frac{1}{2}$ -life is constant which is characteristic of a 1. order reaction. You could also plot $\log([I^-])$ versus times and note that you get a straight line. Thus, the reaction is 1. order.

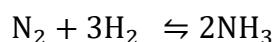
Q14

The compound contains 2 carboxy groups, 1 ether, 1 amine, and one hydroxyl. There is one carbon with 4 different substituents. Thus

“Ether, hydroxyl, carboxyl, amin. The compound is chiral” is correct.

Q15

The balanced reaction equation is



Thus, the partial pressure of nitrogen is lowered:

$$p_{N_2,eq} = P_{N_2,initial} - \frac{1}{2}p_{NH_3} = 52.5bar$$

Likewise

$$p_{H_2,eq} = P_{H_2,initial} - \frac{3}{2}p_{NH_3} = 157.5bar$$

Thus, the equilibrium constant is

$$K_{eq} = \frac{[NH_3]^2}{[N_2][NH_3]^3} = 4.40 \times 10^{-5} bar^{-2}$$

Q16

The repeat unit in PET has the sum formula $C_{10}H_8O_4$. $M_{repeat}=192g/mol$. $n = \frac{\frac{100kg}{mol}}{192g/mol} = 521$

Q17

19.6 g of H_3PO_4 is 0.2 mol. 10 g NaOH is 0.25 mol. Thus, the solution is equivalent to a mixture of 0.15 mol $H_2PO_4^-$ and 0.05 mol HPO_4^{2-} . $pH = pK_2 + \log_{10} \left(\frac{0.05}{0.15} \right) = 6.73$

Q18

You get 2 OH^- for each Mg^{2+} . Thus

$$K_{Mg(OH)_2} = \frac{1}{2}[OH^-]^3 \Rightarrow [OH^-] = \left(2K_{Mg(OH)_2} \right)^{\frac{1}{3}} = 2.24 \times 10^{-4}$$

$$\Rightarrow \text{pH} = 14 - \text{pOH} = 14 - 3.65 = 10.35$$

Q19

The concentration cell:

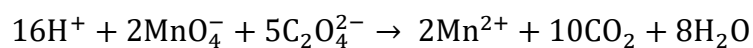
$$E_{\text{cell}} = E_{\text{cathode}}^0 - E_{\text{anode}}^0 + \frac{59.2\text{mV}}{n} \log_{10} \left(\frac{[\text{Ni}^{2+}]_{\text{cathode}}}{[\text{Ni}^{2+}]_{\text{anode}}} \right) = 0 + 29,6\text{mV} \times \log_{10} \left(\frac{0.5}{3 \times 10^{-5}} \right) = 125\text{mV}.$$

As a check for the sign of the solution recall

$$\Delta G = -nFE$$

Thus, for ΔG to be negative (spontaneous process) E has to be positive.

Q20:



Sum of coefficients: 43. If you exclude H^+ and H_2O the sum is 19 which is accepted as an answer as well.